The Stock Market, the Theory of Rational Expectations, and the Efficient Market Hypothesis

Chapter 7

Econ 351 – Monetary Economics

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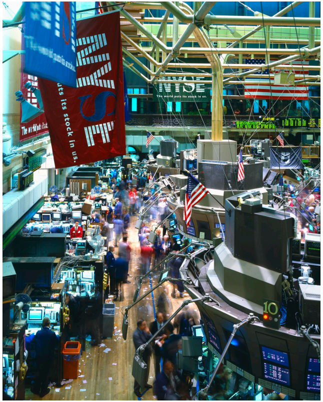
Preview

In this chapter, we introduce **rational expectations** and show how it leads to the **efficient market hypothesis**, with implications for financial markets beyond just stocks.

- A stock is worth the present value of its expected future dividends.
- Expectations are formed rationally, using all available information.
- Hence prices reflect all available information: the efficient market hypothesis (EMH).



The trading floor of the New York Stock Exchange.

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Outline

Asset Pricing Methods

Three Main Approaches

- **Cost approach**
  - Based on historical cost
  - Accounting looks backward; finance looks forward
- **Market approach**
  - Based on comparable market prices
  - Arbitrage-based pricing logic
- **Income approach**
  - Present value of expected future cash flows
  - Discounted cash flow (DCF) method

Why is valuing tech startups especially difficult?

Equity Valuation: Discounted Cash Flow (DCF)

Core Principle

The value of any financial asset equals the present value of its expected future cash flows.

Step 1: Forecast Cash Flows

- Use actual cash flows, not accounting profits.
- Equity cash flows are typically more volatile than debt cash flows.

Step 2: Choose a Discount Rate

- Discount rate = risk-free rate + risk premium.

The One-Period Valuation Model

One-Period Valuation Model

$$P_0 = \frac{D_1}{1 + k_e} + \frac{P_1}{1 + k_e}$$

Interpretation

- Stock price equals the present value of future cash flows.
- Cash flows: dividend and future selling price.

Definitions

- P_0: current stock price
- D_1: dividend at end of year 1
- P_1: stock price at end of year 1
- k_e: required return on equity

The Generalized Dividend Valuation Model

Generalized Dividend Valuation Model

$$P_0 = \frac{D_1}{(1+k_e)} + \frac{D_2}{(1+k_e)^2} + \cdots + \frac{D_n + P_n}{(1+k_e)^n}$$

Key Result

- If P_n is far in the future, its present value is negligible.
- Stock price equals the present value of future dividends.

Dividend Discount Model (DDM)

$$P_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+k_e)^t}$$

The Gordon Growth Model

Gordon Growth Model

$$P_0 = \frac{D_1}{k_e - g} = \frac{D_0(1+g)}{k_e - g}$$

Definitions

- D_0: most recent dividend
- D_1: dividend next period
- g: constant growth rate of dividends
- k_e: required return on equity

Assumptions

- Dividends grow at a constant rate forever.
- $g < k_e$

Outline

Pricing Common Stock

New Information and Stock Prices

Adaptive and Rational Expectations

Arbitrage and the EMH

EMH and Financial Markets

How the Market Sets Stock Prices

Key Idea

- The stock price is set by the buyer willing to pay the highest price.
- This buyer is the one who can make the best use of the asset.

Role of Information

- Better information reduces perceived risk.
- Lower perceived risk increases value.

Stock market parallel: the best-informed investor sets the price, and new information shifts demand.

How New Information Changes Prices

Valuation

- Depends on expectations of future dividends and growth.
- New information changes investors' expectations and therefore stock prices.

Good vs. Bad News

- Good news $\rightarrow$ higher expected cash flows $\rightarrow$ higher stock prices
- Bad news $\rightarrow$ higher risk or lower growth $\rightarrow$ lower stock prices

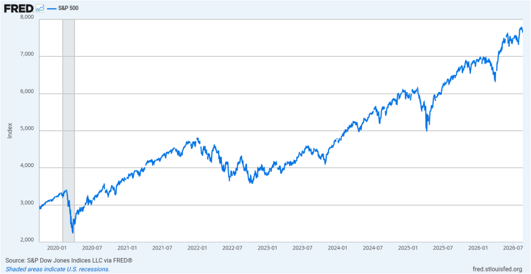
Why Do Stock Prices Change So Often?

Constant Arrival of Information

- New information arrives continuously.
- Expectations about dividends and risk change.

Efficient Market Hypothesis (EMH)

- Prices quickly reflect public information.
- It is hard to consistently beat the market.



The S&P 500 since late 2019 moves every day as news arrives.

Source: FRED, Federal Reserve Bank of St. Louis

Application: Monetary Policy and Stock Prices

Why Does the Fed Affect Stock Prices?

- Monetary policy affects the required return and growth.

Using the Gordon Growth Model

- Lower interest rates $\rightarrow$ lower k_e $\rightarrow$ higher stock prices
- Lower interest rates $\rightarrow$ higher growth (g) $\rightarrow$ higher stock prices

$$P_0 = \frac{D_1}{k_e - g}$$

Anything that lowers k_e or raises g shrinks the denominator and raises the price.

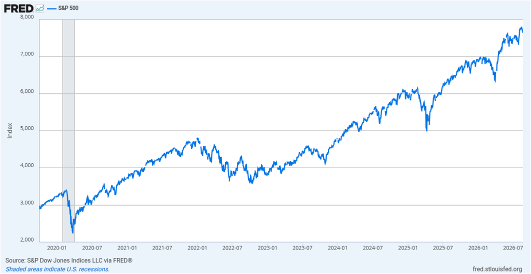
Application: Coronavirus Stock Market Crash (2020)

What Happened?

- The Dow Jones fell about 37% in one month.
- One of the steepest crashes in history.

Why Did Prices Fall?

- Lower expected growth ($g \downarrow$)
- Higher required return ($k_e \uparrow$)



The S&P 500 collapsed in February–March 2020 (shaded recession) and then recovered.

Source: FRED, Federal Reserve Bank of St. Louis

How FinTech Increases Valuation

Economies of Scale & Lower Transaction Costs

- Digital platforms reduce search and transaction costs.
- Technology allows firms to scale quickly at low marginal cost.
- Lower costs $\rightarrow$ higher expected profits $\rightarrow$ higher valuation.

Reducing Information Asymmetry

- Simply connecting borrowers and lenders does not create value.
- Value is created when platforms reduce information asymmetry.
- Better information lowers risk and improves matching efficiency.
- If no informational improvement occurs, long-run value creation is limited.

Outline

Two Theories of Expectation Formation

Two Main Theories of Expectation Formation

- **Adaptive expectations**
  - Expectations based mainly on past data
  - Adjust slowly over time
- **Rational expectations**
  - Expectations use all available information
  - Adjust quickly when new information arrives

Adaptive expectations imply people can be systematically fooled (e.g., by inflation); rational expectations imply forecast errors are random, not systematic.

Adaptive Expectations

- Expectations are based on past inflation.
- People adjust their expectations gradually over time.

$$\pi_t^e = (1 - \lambda) \sum_{j=0}^{\infty} \lambda^j \pi_{t-j}, \qquad 0 < \lambda < 1$$

- Expected inflation is a weighted average of past inflation.
- More recent inflation receives greater weight.

Implications

- Slow adjustment to new information.
- Persistent inflation can raise expected inflation.
- If policy changes quickly enough, the central bank may temporarily surprise the public, creating short-run effects on output and employment.

Rational Expectations

Why Rational Expectations?

- Adaptive expectations adjust too slowly.
- They imply people can be systematically fooled (e.g., by inflation).

How Rational Expectations Work

- People use all available information.
- Expectations equal the best possible forecast.
- Policy or economic changes affect expectations immediately.

Implications

- Expectations adjust quickly.
- Anticipated policy has limited real effects.
- Forecast errors are random, not systematic.

Example: Does Persistent Inflation Stimulate the Economy?

Adaptive Expectations

- Expansionary policy can temporarily boost output.
- But inflation must keep rising.
- To remain effective, inflation must accelerate over time.

Rational Expectations

- Policy is anticipated.
- People adjust wages and prices immediately.
- No systematic real effects — only higher inflation.

Implications for Financial Markets

- Stock prices reflect all available information.
- New information leads to immediate price changes.
- Unexpected events move markets quickly.
- This supports the efficient market hypothesis (EMH).

Rational expectations in financial markets = the efficient market hypothesis: the expected return on a security equals the optimal forecast of its return using all available information.

Which Theory Best Describes Reality?

Adaptive Expectations Work Well When

- Information is limited or difficult to interpret.
- People rely mainly on past trends.

Rational Expectations Work Well When

- People actively process information.
- Markets react quickly to new data.

Bottom Line

- Modern models favor rational expectations.
- Behavioral factors keep adaptive expectations relevant.

Outline

Efficient Market Hypothesis (EMH): Returns

Rate of Return

$$R = \frac{P_{t+1} - P_t + C}{P_t}$$

- R = rate of return
- P_t = price today
- P_{t+1} = price next period
- C = dividend or coupon

Expected Return

$$R^e = \frac{P^e_{t+1} - P_t + C}{P_t}$$

What Does EMH Mean?

Key Assumption

- Expectations are optimal forecasts using all available information.
- Supply and demand imply that the expected return equals the equilibrium return:

$$R^e = R^*$$

Meaning

- Prices fully reflect all available information.
- Current prices are set so that expected returns equal equilibrium returns.

Rationale Behind EMH

Arbitrage Logic

$$R^{of} > R^{*} \Rightarrow P_t \uparrow \Rightarrow R^{of} \downarrow$$

$$R^{of} < R^{*} \Rightarrow P_t \downarrow \Rightarrow R^{of} \uparrow$$

Result

$$R^{of} = R^{*}$$

All unexploited profit opportunities are eliminated.

> If a security's optimal-forecast return R^{of} exceeded R^{*}, investors would buy it, pushing its price up and its return down, until the opportunity disappears.

Random Walk and Exchange Rates

Random Walk

- Future movements cannot be predicted.
- The probability of going up equals the probability of going down.

Implication

- EMH implies exchange rates often follow a random walk.

> Expected news is already priced in, so only *unexpected* news moves prices — and unexpected news is, by definition, unpredictable.

Practical Guide to Investing: Are Advisors Helpful?

Key Lessons from EMH

- Prices reflect all public information.
- Investors cannot consistently beat the market.
- Passive investing is often optimal.

Are Investment Advisors Helpful?

- Analyst reports rely on public information, so they are already reflected in prices.
- Evidence: advisors do not consistently outperform the market.
- *Following analysts will not earn excess returns.*

Hot Tips and Good News

The Myth of the "Hot Tip"

- If the tip is public: it is already reflected in the price.
- If the tip is private: others have likely traded before you.
- *Be skeptical of hot tips.*

Do Stock Prices Always Rise with Good News?

- Markets react to surprises, not news itself.
- Expected news is already priced in.
- Only unexpected news moves prices.

Outline

Why EMH Does *Not* Imply Perfect Markets

Strong (Misleading) Interpretation

If markets were *perfectly* efficient, then:

- All securities would be correctly priced.
- Prices would always equal intrinsic values.
- Prices would change only with fundamental news (no bubbles or crashes).

Reality Check

- Financial markets experience bubbles and crashes.
- This suggests efficiency is limited (not "perfect").

Crowds on Wall Street after the crash of October 1929.

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What Do Market Crashes Tell Us?

Black Monday (1987)

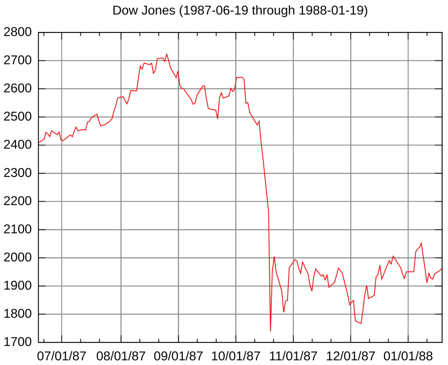
- Oct 19, 1987: the Dow Jones fell over 20% in one day.
- Fundamental explanation: **none clear**.

Tech Bubble Burst (2000–2002)

- The NASDAQ fell about 60% after soaring in the late 1990s.
- Driven by fundamentals? **Not fully**.

Lesson

- Psychology, speculation, and herding can push prices away from intrinsic values.



The Dow Jones around Black Monday, October 19, 1987.

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Market Psychology and Behavioral Finance

Why Do Bubbles & Crashes Occur?

- **Herding:** investors follow the crowd, amplifying swings.
- **Overconfidence:** excessive belief in one's forecasting ability.
- **Fear and panic selling:** losses trigger fire sales during crashes.
- **Irrational exuberance:** over-optimism leads to overvaluation.
- **Liquidity constraints:** forced selling can accelerate declines.

EMH assumes rationality, but real markets reflect behavioral forces.

Are There Unexploited Profit Opportunities?

Does EMH Still Hold? (Forms of EMH)

- **Weak form:** prices reflect past data.
- **Semi-strong form:** prices reflect all public information.
- **Strong form:** prices reflect *all* information (even insider info) — **debatable**.

Critiques

- Bubbles and crashes suggest mispricing.
- Some traders (e.g., hedge funds, insiders) may exploit inefficiencies.

EMH is a useful benchmark, but real markets can contain exploitable inefficiencies.

Chapter summary

- A stock's price is the present value of expected future dividends; with constant growth, $P_0 = D_1/(k_e - g)$ (Gordon growth model).
- New information changes expected dividends, growth, or the required return, and therefore prices; lower interest rates raise stock prices.
- Rational expectations: forecasts use all available information, so forecast errors are random, not systematic.
- EMH: prices reflect all available information; unexploited profit opportunities are eliminated by arbitrage, so hot tips and analyst reports do not earn excess returns.
- Bubbles and crashes (1987, 2000–02) show that markets are not perfectly efficient; behavioral forces matter.